Solved Walkthrough:
From Chief Complaint to SATS Colour

One complaint, every pipeline step

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Abstract

We solve one English chief complaint all the way to a SATS colour and pathway card. Token IDs and the final Softmax $\rightarrow$ Yellow example follow the thesis. Embedding and attention arithmetic use **small illustrative numbers** (true BioBERT uses 768 dimensions and learned weights we do not recompute here).

Contents

1 Setup: what we are solving

Step 0 — The complaint and the goal

Chief complaint X: *"I have a headache and feel feverish. My body aches and I feel weak."*
Goal: follow the pipeline until we get a colour C_{NLP} and a pathway card $P(C)$:

$$X \xrightarrow{g} \tau(X) \to H^{(0)} \xrightarrow{12} h_{[\mathrm{CLS}]} \to \hat{\mathbf{y}} \xrightarrow{f_{\mathrm{SATS}}} C_{\mathrm{NLP}} \to P(C_{\mathrm{NLP}}).$$

Say this

"We will walk one sentence from gate to Yellow pathway, showing numbers at every stage we can."

2 Step 1 — Clinical gate

Before BioBERT runs, score whether X looks clinical.

$$g(X) = \mathrm{clip}\Big(\sum_j s_j(X) - p(X),\, 0,\, 1\Big).$$

Proceed only if $g(X) \geq 0.35$.

Scores for our complaint

Symptom lexicon hits (illustrative, matching the thesis style):

Hit	Score s_j
headache	0.30
feverish	0.25
aches / weak (body)	0.20
$\sum_j s_j$	0.75

Greeting / non-clinical penalty: $p(X) = 0$ (this is a real complaint, not "hello").

$$g(X) = \mathrm{clip}(0.75 - 0,\, 0,\, 1) = 0.75 \geq 0.35.$$

Result for this complaint

Gate decision: PASS. Continue to tokenisation.
(If this had been "Hello, how are you?" with $p(X) = 0.5$ and no symptoms, $g = 0$ and we would stop.)

What just happened

We did not assign a colour yet. We only decided the text is clinical enough for BioBERT.

Say this

"Gate score 0.75 beats the 0.35 threshold, so we proceed."

3 Step 2 — Tokenisation

BioBERT needs integer IDs, not English letters.

$$X \xrightarrow{\mathcal{T}} (t_0, \ldots, t_{M-1}), \qquad M \leq 128.$$

Opening clause (thesis table)

We show the first sentence in detail (as on the Beamer worked-example slide). The second sentence ("My body aches…") is tokenised the same way, then we pad to length 128.

i	Raw	Token	ID
0	n/a	`[CLS]`	101
1	I	i	1045
2	have	have	2031
3	a	a	1037
4–5	headache	head + `##ache`	7994, 8772
6	and	and	1998
7	feel	feel	2514
8–9	feverish	fever + `##ish`	9643, 6804
10	.	.	1012
11	n/a	`[SEP]`	102
12–127	n/a	`<PAD>`	0

- `##` = continuation of a word: "headache" uses *two* positions.
- `[CLS]` will become the summary vector after 12 layers.
- `[SEP]` ends real text; `<PAD>` fills the rest (masked in attention).

What just happened

English string → list of IDs. BioBERT will never see the raw sentence again.

Say this

"We WordPiece-split the complaint, wrap with `[CLS]`/`[SEP]`, and pad to 128 IDs."

4 Step 3 — Embeddings (768-d in the real model)

Each ID becomes a vector by looking up and adding three embeddings:

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0).$$

In BioBERT each of those is length **768**. Below we use a **3-dimensional toy** so you can add by hand.

Toy: token "head" at position $i = 4$

$$\begin{aligned} E_{\text{word}} &= [0.31,\ -0.18,\ 0.52], \\ E_{\text{pos}}(4) &= [0.05,\ 0.02,\ -0.01], \\ E_{\text{seg}}(0) &= [0.00,\ 0.00,\ 0.00], \\ E(t_4) &= [0.36,\ -0.16,\ 0.51]. \end{aligned}$$

Toy: token "fever" at position $i = 8$

$$\begin{aligned}
E_{\text{word}} &= [0.40,\ 0.10,\ -0.20],\\
E_{\text{pos}}(8) &= [0.08,\ -0.01,\ 0.03],\\
E_{\text{seg}}(0) &= [0.00,\ 0.00,\ 0.00],\\
E(t_8) &= [0.48,\ 0.09,\ -0.17].
\end{aligned}$$

Stack into $H^{(0)}$

Every position gets such a vector. Stack as rows:

$$H^{(0)} \in \mathbb{R}^{M\times 768} \quad \text{(real model; our toy would be } M \times 3\text{).}$$

Note

The actual floats come from BioBERT's learned tables (PubMed/PMC + fine-tuning). We do not invent 768 real values here; the toy only shows the *addition rule*.

What just happened

IDs → continuous vectors → matrix $H^{(0)}$ ready for the encoder.

Say this

"Word + position + segment, each 768 long; stack into $H^{(0)}$."

5 Step 4 — Attention (illustrative) and 12 layers

In the real model, each layer builds

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V,$$

then

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \qquad d_k = 64$$

(with 12 heads so $12 \times 64 = 768$).

We cannot multiply the true W_Q here. Instead we **solve a tiny attention problem** with $d_k = 2$ so every number is visible.

Toy setup: `[CLS]` listening to three keys

Suppose the query for `[CLS]` and three keys (symptom-like / filler) are:

$$\begin{aligned}
q &= [1.0,\ 2.0],\\
k_1 &= [1.0,\ 0.0] \quad (\text{"head"-like}),\\
k_2 &= [0.0,\ 1.0] \quad (\text{"fever"-like}),\\
k_3 &= [0.2,\ 0.1] \quad (\text{"I"-like filler}),\\
v_1 &= [1,\ 0], \quad v_2 = [0,\ 1], \quad v_3 = [0,\ 0].
\end{aligned}$$

Compute scores $e_j = (q \cdot k_j)/\sqrt{d_k}$

Here $\sqrt{d_k} = \sqrt{2} \approx 1.414$.

$$\begin{aligned} q \cdot k_1 &= 1.0, & e_1 &= 1.0/1.414 \approx 0.707, \\ q \cdot k_2 &= 2.0, & e_2 &= 2.0/1.414 \approx 1.414, \\ q \cdot k_3 &= 0.4, & e_3 &= 0.4/1.414 \approx 0.283. \end{aligned}$$

Softmax weights α_j

$$\begin{aligned} \exp(e_1) &\approx 2.028, \quad \exp(e_2) \approx 4.114, \quad \exp(e_3) \approx 1.327, \\ \sum \exp(e_j) &\approx 7.469, \\ \alpha_1 &\approx 0.271, \quad \alpha_2 \approx 0.551, \quad \alpha_3 \approx 0.178. \end{aligned}$$

Check: $0.271 + 0.551 + 0.178 = 1$.
`[CLS]` listens most to the "fever"-like key ($\alpha_2 \approx 0.55$), then "head", then filler.

Mix values

$$\text{output} = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \approx 0.271\,[1,0] + 0.551\,[0,1] + 0.178\,[0,0] = [0.271,\ 0.551].$$

What just happened

Scores → weights that sum to 1 → weighted mix of value vectors. That is one attention head on a toy. BioBERT does this with $d_k = 64$, 12 heads, inside each of 12 layers.

After 12 real layers

$$H^{(0)} \to H^{(1)} \to \cdots \to H^{(12)}.$$

Take the first row:

$$h_{[\text{CLS}]} = H^{(12)}_{0,:} \in \mathbb{R}^{768}.$$

Note

We do not write out the true 768 numbers. For the next step we only need: "$h_{[\text{CLS}]}$ is a fixed 768-vector summary of our complaint."

Say this

"Attention: match query to keys, scale, softmax, mix values. Twelve layers later, read `[CLS]`."

6 Step 5 — Softmax classification (thesis numbers)

The five-class head:

$$\mathbf{z} = W h_{[\text{CLS}]} + \mathbf{b}, \qquad \hat{y}_c = \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}.$$

Illustrative logits (from the thesis example)

After the linear map, suppose

$$\mathbf{z} = [-1.2,\ -0.5,\ 1.8,\ 0.3,\ -0.9].$$

Compute exponentials:

Level c	z_c	$\exp(z_c)$ (approx.)
1 (Red)	-1.2	0.301
2 (Orange)	-0.5	0.607
3 (Yellow)	1.8	6.050
4	0.3	1.350
5	-0.9	0.407
Sum		8.715

Probabilities

$$\begin{aligned}
\hat{y}_1 &= 0.301/8.715 \approx 0.035,\\
\hat{y}_2 &= 0.607/8.715 \approx 0.070,\\
\hat{y}_3 &= 6.050/8.715 \approx 0.694,\\
\hat{y}_4 &= 1.350/8.715 \approx 0.155,\\
\hat{y}_5 &= 0.407/8.715 \approx 0.047.
\end{aligned}$$

Check: $0.035 + 0.070 + 0.694 + 0.155 + 0.047 \approx 1.00$.

Chosen level

Largest probability is $\hat{y}_3 \approx 0.69$ (about 72% in rounded deployment talk).

$$\hat{c} = 3.$$

Result for this complaint

For this complaint we take **acuity level 3**.

Say this

"Softmax gives five beliefs that sum to one; we pick the largest—here level 3."

7 Step 6 — Colour map f_{SATS}

$$f_{\mathrm{SATS}}(\hat{c}) = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases}$$

With $\hat{c} = 3$:

$$C_{\mathrm{NLP}} = f_{\mathrm{SATS}}(3) = \textbf{Yellow}.$$

Result for this complaint

Colour for our complaint: Yellow.

What just happened

Integer level $\rightarrow$ hospital colour language.

Say this

"Level three maps to Yellow under f_{SATS}."

8 Step 7 — Pathway card $P(C)$

$$P(C) = \big(T_{\max},\ \text{destination},\ \text{actions},\ \text{escalation}\big).$$

For **Yellow** at KNUST University Hospital:

Field	Value
$T_{\max}$	≈ 60 minutes
Destination	Urgent waiting / OPD urgent (Pediatrics if applicable)
Actions	Nursing vitals/TEWS in parallel, timer, diagnostics as needed
Escalation	Re-assess if timer expires or patient worsens

Result for this complaint

Pathway: Yellow card — be seen within about an hour in the urgent stream; nurse still measures TEWS and may override.

Say this

"Yellow is not only a colour; it is a card with timer, destination, and escalation."

9 Step 8 — Full solution recap

Solved: ***"I have a headache and feel feverish. My body aches and I feel weak."***

1. Gate: $g(X) = 0.75 \geq 0.35 \rightarrow$ proceed.
2. Tokenise: WordPiece IDs with `[CLS]`/`[SEP]`/pad to 128.
3. Embed: word+pos+seg $\rightarrow H^{(0)}$ (768-d in the real model).
4. BioBERT: 12 layers of attention+FFN $\rightarrow h_{[\text{CLS}]}$.
5. Softmax: $\hat{\mathbf{y}} \approx (0.03, 0.07, 0.69, 0.16, 0.05)$, $\hat{c} = 3$.
6. Colour: $C_{\text{NLP}} = \text{Yellow}$.
7. Pathway: $P(\text{Yellow})$ with $\approx 60\,\text{min}$ urgent stream at KNUST University Hospital.

$$X \xrightarrow{g} \cdots \rightarrow \hat{c} = 3 \xrightarrow{f_{\text{SATS}}} \text{Yellow} \rightarrow P(\text{Yellow}).$$

Closing line for the viva

"For this headache-and-fever complaint, the solved pipeline yields Yellow with an urgent pathway card, while the nurse keeps TEWS and override."